

AMENDMENT NO. 2 JULY 2026
TO
IS 16722 : 2018 RADIO FREQUENCY IDENTIFICATION (RFID) SYSTEM FOR
AUTOMOTIVE APPLICATIONS — SPECIFICATION

(Page 1, clause 1) — Substitute the following for the existing:

‘1 SCOPE

This standard covers the general definitions and abbreviations which are common for RFID used in automotive applications and the different applications of RFID including electronic toll collection (ETC), parking fee collection (PFC) and vehicle identification, etc.

The standard highlights the basic architecture and test requirements of the radio frequency identification (RFID) tags and transceivers for electronic toll collection ‘ETC’ and tractor trailer interchangeability application.’

(Page 1, clause 3.16) — Insert the following at the end:

‘g) TID — Tag identifier.’

(Page 2, clause 4.1) — Substitute the following for the existing:

‘4.1 Power

The tags in ETC and tractor trailer interchangeability application are passive.’

(Page 2, clause 4.5, heading) — Substitute the following for the existing:

‘4.5 Dimensions (Including the Substrate/Backing) (For ETC Application Only)’

[Page 2, clause 4.6, (see also Amendment No. 1)] — Substitute the following for the existing:

‘4.6 Physical Printing of Tag ID on the Tag for ETC Application

The tag ID/EPC ID shall be physically printed on the front adhesive side of the tag using the hexadecimal numbering and QR code system as per ISO/IEC 18004 : 2015 standard and shall be adequately clear for easy visual recognition. Printing and design should be as defined by NHAI from time to time.’

[Page 2, clause 4.7, (see also Amendment No. 1)] — Substitute the following for the existing:

‘4.7 TAG Position and Orientation

4.7.1 For ETC Application

Regarding fixation of RFID tag prevailing regulations as may be decided by Government of India shall be followed.

4.7.2 For Tractor Trailer Interchangeability Application

The RFID tags shall be fitted on an arc with a tentative radius of 1 300 mm to 1 900 mm from the kingpin as its centre.’

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[Page 2, clause **4.8**, (see also Amendment No. 1)] — Substitute the following for the existing:

‘4.8 Tamper Proof RFID Label

4.8.1 For ETC Application

The tags should be RFID tamper proof label specially designed for tagging directly to a surface, such as glass (windshield) of an automobile. Any attempt to rip or tamper the label (tag) should result in disabling the functionality of the tags to ensure a unique one to one relationship between the tag and vehicle thereby preventing unauthorized tag removal and transfers. Such features of the RFID label should result in following actions:

- a) Destroy damage the antenna; and
- b) Break the chip-antenna connection.

4.8.2 For Tractor Trailer Interchangeability Application

The RFID tag shall be tamper-proof and designed for direct installation on the trailer. Any attempt at removal or tampering shall permanently render the tag non-functional, thereby ensuring a unique one-to-one association with the trailer and preventing unauthorized transfer of tags. Physical tampering shall result in destruction or damage of the chip and breakage of the internal connection.’

[Page 2, clause **4.9**, (see also Amendment No. 1)] — Substitute the following for the existing:

‘4.9 Memory

4.9.1 For ETC Application

The minimum unique tag ID memory size should be 96 bits, minimum EPC memory size should be 96 bits and minimum user memory should be 512 bits.

4.9.2 For Tractor Trailer Interchangeability Application

Memory the minimum EPC memory size should be 96 bits and maximum user memory shall be up to 688 bits.’

[Page 2, clause **4.10**, (see also Amendment No. 1)] — Substitute the following for the existing:

‘4.10 Range

4.10.1 For ETC Application

6 m (min) RFID tag fixed on vehicle windshield from inside in open air.

4.10.2 For Tractor Trailer Interchangeability Application

It should be readable within a range from 250 mm to 1 100 mm.’

[Page 2, clause **4.12**, (see also Amendment No. 1)] — Substitute the following for the existing:

‘4.12 Tag Antenna

4.12.1 For ETC Application

The tag antenna shall be printed or etched.

4.12.2 For Tractor Trailer Interchangeability Application

The tag antenna polarization shall be linear or circular.’

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(Page 2, clause **4.14**, heading) — Substitute the following for the existing:

‘4.14 Tag Architecture and Contents (For ETC Application Only)’

(Page 3, clause **4.15**, (see also Amendment No. 1)] — Substitute the following for the existing:

‘4.15 Additional Specification of RFID Tags (For Tractor Trailer Interchangeability Application Only)’

4.15.1 Chip Type

The chip type used in the RFID tag shall be UCODE 8/UCODE 7/MR6/M4QT/H9/IMPINJ.

4.15.2 Re-write Capability

The tag re-writable capability shall be around 100 000 cycles.

4.15.3 Tag Data

The tag shall carry data embedded with all parameters given below:

<i>Sl No.</i>	<i>Field</i>	<i>Description</i>
(1)	(2)	(3)
i)	Tag ID	Unique EPC/TID identifier of the tag
ii)	Serial number	OEM serialized tag identifier
iii)	TA certification number	Type approval reference
iv)	GCW of trailer	Gross combined weight
v)	VIN	Vehicle identification number of the asset
vi)	Axle count	Number of axles associated
vii)	Registration number	Registered vehicle ID*

4.15.4 Minimum Number of Tags

The minimum number of tags mounted on the trailer shall not be less than 5. Additional number of tags can be provided to ensure operational continuity.’

(Page 3, clause **4.15**) — Insert the following at the end:

‘4.16 Classification of Tests

4.16.1 Criteria for Approval

Minimum 20 samples of RFID tags shall be submitted for testing together with the relevant standards, for ETC and tractor trailer interchangeability applications.

In case of failure in one or more type tests, the testing authority may call for fresh samples not exceeding twice the number of original samples and subject them to test(s) in which failure occurred. If, in repeated test(s) no failure occurs, the test may be considered to have been satisfactory.

Curing time for the tag will be 15 min for test mentioned **4.16.2** to **4.16.6** and 24 h for the rest of the tests unless mentioned specifically in the test.

*RFID tags shall be installed on trailers by the trailer manufacturer and parameters 1 to 6 above shall be encoded in the tag prior to type approval or sale. Registration certificate (RC) shall be updated through a KYC-based process (similar to the existing ETC/FASTag system), at the time of delivery to the operator after sale.

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4.16.2 Visual Examination

4.16.2.1 For ETC application

The maximum dimension will be checked for 50 cm² with no tolerance. Printing of the hexadecimal TID should be easily visible.

4.16.2.2 For tractor trailer interchangeability application

The RFID tag shall show no visible signs of cracking, deformation, delamination, corrosion or physical damage and shall maintain its mechanical and structural integrity

4.16.3 Functional Test

The tag would be read/ tested with a transceiver for the functional usage and the EPC/TID will be noted for further tests.

4.16.4 Range Test

4.16.4.1 For ETC application

The tags will be tested by an approved transceiver inside an anechoic chamber for a distance of 6 m. Functional test will be conducted at this range.

4.16.4.2 For tractor trailer interchangeability application

The tags will be tested by an approved transceiver inside an anechoic chamber for within a range from 250 mm to 1 100 mm. Functional test will be conducted at this range.

4.16.5 Interoperability Test

4.16.5.1 For ETC application

The tag would be functionally tested with one or more than one approved fixed transceivers and one or more than one handheld transceivers for interoperability.

4.16.5.2 For tractor trailer interchangeability application

The tag would be functionally tested with one or more than one approved transceiver for interoperability.

4.16.6 Speed Test (For ETC Application Only)

The tag would be functionally tested with an approved reader at a speed of minimum speed of 20 kmph to maximum speed of 80 kmph.

4.16.7 Electrostatic Discharge Test

The system shall be subjected to tests as per ISO 10605. The maximum test signal severity levels shall be ± 3 kV at unpowered mode.

4.16.8 Endurance Test

4.16.8.1 For ETC application

The RFID tag and transceiver shall be suitably mounted at an ambient temperature. The position of tag and transceiver shall be fixed representing the actual mounting at a distance specified in **4.10**. A shutter arrangement would be made so that the shutter moves back and forth between the line of vision of tag and transceiver. The shutter shall be operated for 100 000 cycles. Shutter on time to be 3 s and off time to be 2 s that is, 5 s constitute one cycle. The transceiver should read the tag properly.

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4.16.8.2 *For tractor trailer interchangeability application*

The RFID tag and transceiver shall be suitably mounted at an ambient temperature. The position of tag and transceiver shall be fixed representing the actual mounting within a range from 250 mm to 1 100 mm. A shutter arrangement would be made so that the shutter moves back and forth between the line of vision of tag and transceiver. The shutter shall be operated for 100 000 cycles. Shutter on time to be 3 s and off time to be 2 s that is, 5 s constitute one cycle. The transceiver should read the tag properly.

4.16.9 *Vibration Test*

4.16.9.1 *For ETC application*

The test specimen (TAGS) mounted on a wind shield glass shall be rigidly fixed on a suitable vibrating machine constructed to produce simple harmonic function (a total amplitude of 1.5 mm) and shall be subjected to vibration through a frequency range of 10-55-10 Hz in a sweep period of 1 min with continuously varying frequencies. The vibration shall be applied for not less than 1 h in the directions of each of the 3 major axes of the light.

The tag after vibration test shall meet the requirements of visual examination and functional test.

4.16.9.2 *For tractor trailer interchangeability application*

The test specimen shall be rigidly fixed on a suitable vibrating machine constructed to produce simple harmonic function (a total amplitude of 1.5 mm) and shall be subjected to vibration through a frequency range of 10-55-10 Hz in a sweep period of 1 min with continuously varying frequencies. The vibration shall be applied for not less than 1 h in the directions of each of the 3 major axes of the light.

The tag after vibration test shall meet the requirements of visual examination and functional test.

4.16.10 *Damp Heat Test*

The RFID should be tested according to IS 9000 (Part 5/Sec 2). The test is carried out at + 25° to + 55 °C, humidity 95 percent. 6 cycles (each test cycle of 24 h) should be run with the system in off condition.

The functional test will be conducted on the TAG at start of 2nd, 4th, 6th cycle and after the test.

4.16.11 *Ingress Test*

4.16.11.1 *For ETC application*

The ingress test shall be carried out in accordance with IS/IEC 60947-1 in conjunction with IS/IEC 60529 for IP 54 compliance.

The functional test will be conducted on the TAG before, during and after the test.

4.16.11.2 *For tractor trailer interchangeability application*

The ingress test shall be carried out in accordance with IS/IEC 60947-1 in conjunction with IS/IEC 60529 for IP 65 compliance.

The functional test will be conducted on the TAG before and after the test.

4.16.12 *Corrosion Resistance Test*

The system should be tested according to 4.8 of IS 10250 for 96 h.

The functional test will be conducted on the TAG before, during and after the test.

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4.16.13 Operating Temperature Test

4.16.13.1 For ETC application

The sample tags (3 numbers) shall be pasted on a piece of vehicle windshield glass and shall be placed in cyclic chamber and subjected to operating temperature test as per specifications mentioned below:

<i>Sl No.</i>	<i>Parameters</i>	<i>Specification</i>
(1)	(2)	(3)
i)	Temperature	+ 80 °C and – 20 °C
ii)	Duration	16 h at each temperature
iii)	No. of cycles	1

The tag after the test shall meet the requirements of visual examination and functional test.

NOTE — The operating temperature test shall be in accordance with IS 9000 (Part 2) and (Part 3).

4.16.13.2 For tractor trailer interchangeability application

The sample tags (3 numbers) shall be pasted on a piece of metal sheet (similar to trailer body structure) and shall be placed in cyclic chamber and subjected to operating temperature test. The test shall be carried at two temperatures, + 80 °C and – 20 °C, with a duration of 16 h at each temperature. The tag after the test shall meet the requirements of visual examination and functional test.

4.16.14 Storage Temperature Test

The system shall be placed in cyclic chamber and subjected to storage temperature test as per specifications mentioned below:

<i>Sl No.</i>	<i>Parameters</i>	<i>Specification</i>
(1)	(2)	(3)
i)	Temperature	+ 100 °C and – 40 °C
ii)	Duration	16 h at each temperature
iii)	No. of cycles	1

The tag after the test shall meet the requirements of visual examination and functional test.

NOTE — The storage temperature test shall be in accordance with IS 9000 (Part 2) and (Part 3).

4.16.15 UV Radiation Test

The system shall be exposed to radiation from a source having a spectral energy distribution similar to that of a black body at a temperature between 5 500 K and 6 000 K. Appropriate filters shall be placed between the source and the samples so as to reduce as far as possible radiations with wave lengths smaller than 295 nm and greater than 2 500 nm. The samples shall be exposed to an energetic illumination of $1\,200\text{ W/m}^2 \pm 200\text{ W/m}^2$ for a period such that the luminous energy that they receive is equal to $4\,500\text{ MJ/m}^2 \pm 200\text{ MJ/m}^2$. Within the enclosure, the temperature measured on the black panel placed on a level with the samples shall be $50\text{ °C} \pm 5\text{ °C}$. In order to ensure a regular exposure, the samples shall revolve around the source of radiation at a speed between 1 litre/min and 5 litre/min. The samples shall be sprayed with distilled water of conductivity lower than 1 mS/m at a temperature of $23\text{ °C} \pm 5\text{ °C}$, in accordance with the following cycle:

- Spraying — 5 min; and
- Drying — 25 min.

The tag after the test shall meet the requirements of visual examination and functional test.

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4.16.16 Peel Strength Test (For ETC Application and Tractor Trailer Interchangeability Application Pertaining to Adhesive Type RFID Tags Only)

The dimensions for the test specimen for ETC application shall be 25 mm wide and 300 mm length approximately. For tractor trailer interchangeability application, the size of the test specimen may be as per the declaration of the manufacturer. A strip of tape is applied to standard test panel or other surface of interest with controlled pressure. Curing time for the tag will be 72 h. The tape is peeled from the panel at 180° angle at a 200 mm/min rate, during which time the force required to effect peel is measured.

Acceptance criteria is as per the specifications below:

<i>Sl No.</i>	<i>Property</i>	<i>Requirement</i>
(1)	(2)	(3)
i)	Peel strength initial after 20 min at room temperature	Minimum 800 g/2.5 cm
ii)	Peel strength after 48 h at room temperature	Minimum 1 200 g/2.5 cm
iii)	Peel strength after environmental test	Minimum 1 200 g/2.5 cm
NOTE — See ASTM D 3330.		

4.16.17 Acceptance Tests For Tags

The following shall constitute acceptance tests for tags after every 6 months or 100 000 samples, whichever achieved later.

4.16.17.1 For ETC Application

- a) Visual examination (see 4.16.2);
- b) Functional test (see 4.16.3);
- c) Interoperability (see 4.16.5); and
- d) Peel strength test (see 4.16.16).

4.16.17.2 For Tractor Trailer Interchangeability Application

- a) Visual examination (see 4.16.2);
- b) Functional test (see 4.16.3);
- c) Range test (see 4.16.4);
- d) Interoperability (see 4.16.5);
- e) Ingress test (see 4.16.11); and
- f) Storage temperature test (see 4.16.14).

[Page 5, clause 5, (see also Amendment No. 1)] — Substitute the following for the existing:

‘5 TRANSCIVERS (READ/WRITE UNIT)

5.1 Fixed Transceivers

5.1.1 Transmission Frequency

The operating frequency range should be between 865 MHz and 868 MHz.

5.1.2 Communication

Ethernet/serial — USB communication (EIA standard RS 232/RS 485) and any other option.

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5.1.3 RF Power Maximum (For ETC Application Only)

1 W — Transmitted/conducted and 4 W — EIRP (equivalent isotropically radiated power) as per WPC norms.

5.1.4 Range

5.1.4.1 For ETC application

The transceiver should be able to read the tag within the range of 4 m to 8 m. Test to be conducted inside an anechoic chamber.

5.1.4.2 For tractor trailer interchangeability application

The transceiver should be able to read the tag within the range from 250 mm to 1 100 mm. Test to be conducted inside an anechoic chamber.

5.1.5 Protocol

5.1.5.1 For ETC application

The protocol to be used is EPC Gen 2, ISO 18000-6C/3.

5.1.5.2 For tractor trailer interchangeability application

The protocol to be used is EPC global Class-1 Gen-2/ISO 18000-6C.

5.1.6 Data Transmission Rate

The data transfer rate of the tags should be minimum 100 kbps.

5.1.7 Polarization

5.1.7.1 For ETC application

The polarization can be circular or linear, based on individual toll plaza designs, distance between 2 lanes, reader, tag mountings.

5.1.7.2 For tractor trailer interchangeability application

The polarization shall be linear or circular.

5.1.8 Crosstalk

The communication should be limited to a transceiver and a tag.

5.1.9 External Antenna Ports

5.1.9.1 For ETC application

The system should have a minimum 1 internal/external antenna.

5.1.9.2 For tractor trailer interchangeability application

As per system requirements.

5.2 Handheld Transceivers (For ETC Application Only)

The handheld RFID reader should be a single integrated module including handheld unit and RFID module. The operating transmission frequency range should be between 865 MHz and 868 MHz.

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5.2.1 Readability/Writability

The handheld wireless readers should be able to read the tag within a range of 0.5 m and 2 m.

5.2.2 Memory

A non-volatile memory of minimum 512 MB.

5.2.3 Communication

Communication shall be possible using wireless, Bluetooth, USB, RS232, Ethernet, etc.

5.2.4 Display

Visual indication should be present.

5.2.5 Power Supply

The power supply should be a Li-ion battery with a minimum 4 h back up with transmitter 'ON' and 24 h standby mode.

5.3 ETC POS Reader (For ETC Application Only)

The ETC POS reader should be a single integrated module including antenna and RFID module.

5.3.1 Transmission Frequency

The operating frequency range should be between 865 MHz and 868 MHz.

5.3.2 Readability/Writability

The handheld wireless readers should be able to read the tag within a range of 0 inch to 6 inch.

5.3.3 Memory

A non-volatile memory of minimum 512 MB.

5.3.4 Communication

Communication shall be possible using wireless, Bluetooth, USB, RS232, Ethernet, etc.

5.3.5 Display

Visual indication should be present.

5.3.6 Power Supply

The power supply should be a Li-ion battery with a minimum 4 h back up with transmitter 'ON' and 24 h standby mode.

5.4 Power Supply (For Tractor Trailer Interchangeability Application Only)

As per system requirements.

5.5 Display (For Tractor Trailer Interchangeability Application Only)

Visual Indication should be present (for power, VLTD communication or connectivity and tag reading status, etc).

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5.6 Transceivers Position and Orientation (For Tractor Trailer Interchangeability Application Only)

Transceivers shall be fitted on at a distance of 1 000 mm to 2 200 mm from the centre point of the 5th wheel gear on the centre line of the vehicle.'

(Page 6, clause 6.5) — Substitute the following for the existing:

'6.5 EMI/EMC Test

6.5.1 For ETC Application

For the DUT shall be subjected to following tests:

- a) Measurement of radiated electromagnetic emissions from the system according to CISPR 22 [see IS 6874 (Part 7)]; and
- b) Method(s) of testing for immunity of the system to electromagnetic radiation according to IEC 61000-4-3 [see IS 14700 (Part 4/Sec 3)] up to 1 000 MHz with 10 V/m severity level.

No deviation in DUT functionality shall be observed during this test. DUT shall meet criterion A.

6.5.2 For Tractor Trailer Interchangeability Application

For automotive application, compliance with EMI/EMC requirements in accordance with AIS-004 (Part 3) shall be applicable.

Acceptance Criteria — Test results shall comply with the requirement specified in AIS-004 (Part 3).'

(Page 7, clause 6.9, heading) — Substitute the following for the existing:

'6.9 Ingress Test (For ETC Application and Tractor Trailer Interchangeability Application as per Transceiver Specification)'

(Page 7, clause 6.14) — Substitute the following for the existing:

'6.14 Acceptance Tests for Transceivers

The following shall constitute acceptance tests for transceivers after every one year or 10 000 samples, whichever is achieved earlier:

6.14.1 For ETC Application

- a) Visual examination (see 6.2);
- b) Measurement of dimensions; and
- c) Interoperability tests (see 6.3).

6.14.2 For Tractor Trailer Interchangeability Application

- a) Visual examination (see 6.2);
- b) Interoperability tests (see 6.3);
- c) Performance test (see 6.4); and
- d) Ingress test (see 6.9).'

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